

Linux RAG Dataset - PDF 5: Process & System Management

ps

Category: Process Management

Description: Display running processes

Syntax: ps [options]

Example: ps aux

Notes: Shows active processes.

Related Commands: top

Difficulty: Beginner

top

Category: Monitoring

Description: Monitor system processes in real time

Syntax: top

Example: top

Notes: Displays CPU and memory usage.

Related Commands: htop

Difficulty: Beginner

htop

Category: Monitoring

Description: Interactive process viewer

Syntax: htop

Example: htop

Notes: Enhanced version of top.

Related Commands: top

Difficulty: Beginner

kill

Category: Process Management

Description: Terminate a process using PID

Syntax: kill PID

Example: kill 1234

Notes: Sends SIGTERM by default.

Related Commands: pkill

Difficulty: Beginner

pkill

Category: Process Management

Description: Kill processes by name

Syntax: pkill process_name

Example: pkill firefox

Notes: Uses process names.

Related Commands: kill

Difficulty: Intermediate

killall

Category: Process Management

Description: Kill all processes with same name

Syntax: killall process_name

Example: killall chrome

Notes: Terminates all matching processes.

Related Commands: pkill

Difficulty: Intermediate

nice

Category: Performance

Description: Start a process with priority

Syntax: nice -n VALUE COMMAND

Example: nice -n 10 python app.py

Notes: Lower priority number = higher priority.

Related Commands: renice

Difficulty: Intermediate

renice

Category: Performance

Description: Change running process priority

Syntax: renice VALUE PID

Example: renice 5 1234

Notes: Modify priority after start.

Related Commands: nice

Difficulty: Intermediate

systemctl

Category: Services

Description: Manage systemd services

Syntax: systemctl COMMAND SERVICE

Example: systemctl restart ssh

Notes: Used in modern Linux distributions.

Related Commands: service

Difficulty: Intermediate

service

Category: Services

Description: Manage services (legacy)

Syntax: service NAME ACTION

Example: service apache2 restart

Notes: Legacy service manager.

Related Commands: systemctl

Difficulty: Intermediate

journalctl

Category: Logs

Description: View systemd logs

Syntax: journalctl [options]

Example: journalctl -xe

Notes: Reads journal logs.

Related Commands: dmesg

Difficulty: Intermediate

dmesg

Category: Logs

Description: Display kernel messages

Syntax: dmesg

Example: `dmesg | tail`

Notes: Useful for hardware issues.

Related Commands: `journalctl`

Difficulty: Intermediate

free

Category: Memory

Description: Display memory usage

Syntax: `free -h`

Example: `free -h`

Notes: Shows RAM and swap usage.

Related Commands: `vmstat`

Difficulty: Beginner

uptime

Category: System Info

Description: Show system uptime

Syntax: `uptime`

Example: `uptime`

Notes: Shows load averages.

Related Commands: `w`

Difficulty: Beginner

vmstat

Category: Performance

Description: Report virtual memory statistics

Syntax: `vmstat`

Example: `vmstat 2`

Notes: Useful for performance monitoring.

Related Commands: `free`

Difficulty: Intermediate

iostat

Category: Disk

Description: Display CPU and disk statistics

Syntax: `iostat`

Example: `iostat -x`

Notes: Requires `sysstat` package.

Related Commands: `vmstat`

Difficulty: Intermediate

sar

Category: Performance

Description: Collect system activity information

Syntax: `sar`

Example: `sar -u`

Notes: Historical performance reports.

Related Commands: `iostat`

Difficulty: Advanced

lsof

Category: Files

Description: List open files

Syntax: lsof

Example: lsof -i :80

Notes: Shows which process opened a file.

Related Commands: fuser

Difficulty: Intermediate

fuser

Category: Files

Description: Identify process using file

Syntax: fuser FILE

Example: fuser /home/file.txt

Notes: Shows process IDs.

Related Commands: lsof

Difficulty: Intermediate

strace

Category: Debugging

Description: Trace system calls

Syntax: strace COMMAND

Example: strace ls

Notes: Useful for debugging.

Related Commands: ltrace

Difficulty: Advanced

ltrace

Category: Debugging

Description: Trace library calls

Syntax: ltrace COMMAND

Example: ltrace ls

Notes: Library function tracing.

Related Commands: strace

Difficulty: Advanced

crontab

Category: Scheduling

Description: Manage scheduled jobs

Syntax: crontab -e

Example: crontab -l

Notes: Edit cron jobs.

Related Commands: at

Difficulty: Intermediate

at

Category: Scheduling

Description: Run a command once later

Syntax: at TIME

Example: at 10:30

Notes: One-time scheduled task.

Related Commands: crontab

Difficulty: Intermediate

nohup

Category: Background Jobs

Description: Run command after logout

Syntax: nohup COMMAND

Example: nohup python app.py &

Notes: Ignores hangup signal.

Related Commands: &

Difficulty: Intermediate

jobs

Category: Shell

Description: Show background jobs

Syntax: jobs

Example: jobs

Notes: Lists current shell jobs.

Related Commands: bg

Difficulty: Beginner

bg

Category: Shell

Description: Resume job in background

Syntax: bg JOB_ID

Example: bg %1

Notes: Continues stopped jobs.

Related Commands: fg

Difficulty: Beginner

fg

Category: Shell

Description: Bring job to foreground

Syntax: fg JOB_ID

Example: fg %1

Notes: Interact with background job.

Related Commands: bg

Difficulty: Beginner

time

Category: Performance

Description: Measure execution time

Syntax: time COMMAND

Example: time python app.py

Notes: Shows execution duration.

Related Commands: uptime

Difficulty: Beginner

watch

Category: Monitoring

Description: Run command repeatedly

Syntax: watch COMMAND

Example: watch df -h

Notes: Updates output periodically.

Related Commands: top

Difficulty: Beginner

sleep

Category: Shell

Description: Pause execution

Syntax: sleep SECONDS

Example: sleep 5

Notes: Useful in scripts.

Related Commands: wait

Difficulty: Beginner